08·05·2025 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025– April 2025		
Maximum Marks: 50 **Examination: End-Semester Examination**		**Duration: 2 Hrs.**
Programme code: 01 **Programme: B. Tech-** COMP	**Class: SY**	**Semester: IV(SVU 2023)**
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department:** COMP	
Course Code: 216U01C404	**Name of the Course: Operating System**	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Represent with the help of neat diagram how a program written in high level language is executed on a computer. Explain highlighting the role of various system software involved.	05
ii)	Differentiate between Monolithic and Microkernel OS architectures.	05
iii)	What is PCB? Explain the contents of PCB.	05
Q.2	Attempt any **one**	10
i)	a. What are ULTs and KLTs? State the advantages of ULTs over KLTs.	5
	b. Draw 7-state diagram of life cycle of a process and explain the various state transitions.	5
ii)	a. What is process switching? Explain the steps involved in full process switch.	5
	b. Consider the following set of Processes with the length of CPU burst given in ms	5

Process	Arrival time	Burst time
P1	0	3
P2	2	6
P3	4	4
P4	6	5
P5	8	2

Schedule the processes according to Round Robin scheduling (time quantum = 1)
(i) Draw the Gantt chart
(ii) Calculate average turnaround time and average waiting time

Q.3	Attempt any **one**	10
i)	Explain critical section in relation to process concurrency. Demonstrate the solution to bounded buffer Producer-Consumer problem using semaphores with the help of pseudo code.	

ii) Explain the necessary and sufficient conditions for deadlock.
Assume that there are 5 processes, P_0 through P_4, and 4 types of resources-A, B, C, D.
At time T_0 we have the following system state:
Max Instances of Resource Type A = 3
Max Instances of Resource Type B = 17
Max Instances of Resource Type C = 16
Max Instances of Resource Type D = 12

	Allocation Matrix (N0 of the allocated resources By a process)				Max Matrix Max resources that may be used by a process			
	A	B	C	D	A	B	C	D
P_0	0	1	1	0	0	2	1	0
P_1	1	2	3	1	1	6	5	2
P_2	1	3	6	5	2	3	6	6
P_3	0	6	3	2	0	6	5	2
P_4	0	0	1	4	0	6	5	6

Using Banker's algorithm
a. Write the Need Matrix
b. Illustrate that the system is in safe state by demonstrating the sequence in which the processes may complete.

Q.4 Attempt the following **10**

Suppose that a disk drive has 200 cylinders numbered 0 to 199. The drive is currently serving a request at cylinder 53 and the previous request was at cylinder 35. The queue of pending requests in FIFO order is:
{98, 183, 37, 122, 14, 124, 65, 67}.
Starting from the current head position, what is the total distance (in cylinders) that the disk head moves to satisfy all pending requests for the following Disk scheduling algorithms: (i) SSTF (ii) SCAN (iii) C-SCAN (iv) LOOK
Draw the Gant chart to depict the head movement for the above mentioned algorithms.
Which algorithm will result in least average head movement? Justify.

Q.5 Attempt the following **10**

a. Write a note on File Organization **5**

b. Which entity is responsible for run-time mapping of memory from virtual addresses to physical addresses? Explain briefly its functioning. **5**